

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1854

South Meadow 345 kV
June 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1854. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the South Meadow 345 kV substation in the Pennsylvania area. The request was received on August 10, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 494 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,465,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TARPON SPRINGS to LA CANAA 138kV	CEDAR CREEK to YORKANA 345kV	143.3	200	7.7
TARPON SPRINGS to CEDAR CRK SUBSTATION 138kV	CEDAR CRK to YORKANA 345kV	116.7	280	5.9
S MEADOW to MONROE NE 345kV	S MEADOW to POCKET N 345kV	115.7	1,187	63.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TARPON SPRINGS to LA CANAA 138kV	Rebuild line with higher-capacity conductor	\$1,852,000	86% (\$1,595,000)
TARPON SPRINGS to CEDAR CRK 138kV Line	Rebuild line with higher-capacity conductor	\$1,111,000	97% (\$1,081,000)
S MEADOW to MONROE NE 345kV	Rebuild line with higher-capacity conductor	\$13,817,000	100% (\$13,789,000)
	Total allocated to project		\$16,465,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.